

MVIS Railway Digital Twin - Client Visualization Blueprint

This document is for building a Three.js client demo. The audience is non-technical railway officials. Do NOT visualize databases, cloud infrastructure, MQTT, APIs, AI models, storage pipelines, or engineering internals. The experience should feel like a realistic railway inspection portal where visitors can watch what happens when a train passes through the system. The user views the world in third-person mode and can walk around the portal.

Core Story

1. Visitor enters the railway inspection site. 2. The railway track is idle. 3. A train approaches. 4. Portal automatically activates. 5. Cameras and lights wake up. 6. Train enters inspection zone. 7. Components are visually inspected. 8. Defects (if any) are highlighted. 9. Evidence is generated. 10. Alert is shown. 11. Train exits. 12. Portal returns to standby mode.

Environment

Create a realistic railway environment with track, ballast, poles, signals, maintenance pathways, industrial inspection gantry, realistic train models, ambient railway sounds and dynamic lighting. The portal should be the visual hero of the scene.

Third Person Experience

Player walks around using WASD. Camera follows behind the player. Player can inspect cameras, portal, train and control screens. No FPS mode. Use cinematic third-person presentation.

Train Arrival Sequence

When a train is detected, warning lights begin blinking. Portal LEDs glow blue. Scanning beams become visible. The entire portal appears to wake up automatically.

Inspection Sequence

As the train moves through the portal: - Wheels are highlighted. - Springs are highlighted. - Brake assemblies are highlighted. - Bogies are highlighted. Use subtle glowing overlays instead of technical bounding boxes.

Defect Demonstration

Allow a demo mode where one spring becomes defective. The system visually zooms attention to that component. Show a railway-friendly information card: Defect Type Coach Number Severity Inspection Time

Evidence Visualization

After detection, show evidence cards floating beside the train: Inspection Image Inspection Snapshot Defect Snapshot Inspection Report This should look like an inspection process rather than a software workflow.

Control Room View

A nearby building contains a control room. Large screens display: Total Trains Inspected Defects Found Today's Activity Latest Alerts Keep terminology simple.

AutoCAD Integration Notes For Future AI

When the AutoCAD drawing is supplied: - Extract actual portal dimensions. - Position cameras according to drawing. - Match track spacing. - Match inspection zones. - Replace placeholder geometry with CAD-derived geometry.

Codex / SOL Model Prompt

Build an immersive Three.js railway inspection digital twin. Audience is railway officials. Focus on storytelling, realism and visual understanding. The portal should visibly activate when a train arrives. Show inspections happening visually. Show defects clearly. Keep technical architecture hidden. Create cinematic animations, environmental audio, realistic lighting and guided camera moments. The experience should explain the value of the inspection system without requiring technical knowledge.